

1	2	3	4	5	6	7	8
Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Rails: V5_SYS from the 5 V inputs, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. No PCB placement or routing has been performed.							
A							
B	POWER 1 / 5 V in, USB, priority power mux File: power_sources.kicad_sch			BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch			
	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch			BOARD 3 / log flash and microSD File: storage.kicad_sch			
C	POWER 3 / peripheral ports File: power_periph.kicad_sch			BOARD 4 / PWM outputs File: actuators.kicad_sch			
D	BOARD 1 / RP2354B File: mcu.kicad_sch						
E							
F							
1	2	3	4	5	6	7	8

5 V INPUTS ONLY: BEC/boost 4.75–5.5 V (J3), USB VBUS ~5 V (J4). NO RAW CELL INPUT ON THIS BOARD.

A

B

C

D

E

F

A

B

C

D

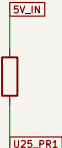
E

F

D23
SMAJ5.0A, 5.0 V standoff uni-directional TVS



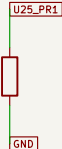
R42
32.4k / 1%, PR1 top (5V_IN)



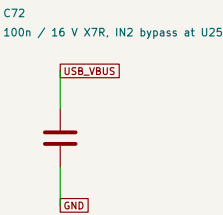
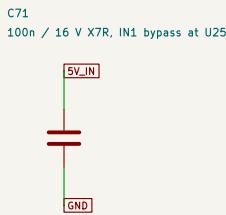
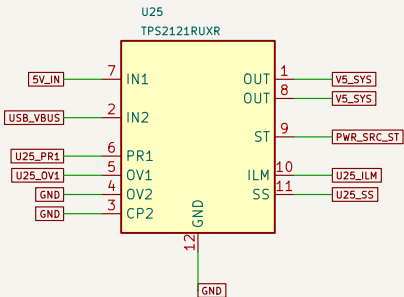
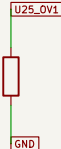
R44
45.3k / 1%, OV1 top (5V_IN)



R43
10k / 1%, PR1 bottom



R45
10k / 1%, OV1 bottom



#FLG1
PWR_FLAG



#FLG2
PWR_FLAG



#FLG3
PWR_FLAG



#FLG4
PWR_FLAG



C16
100n / 16 V X7R, J3 HF bypass



C67
4.7u / 10 V X7R, J3 bulk



C68
4.7u / 10 V X7R, J3 bulk



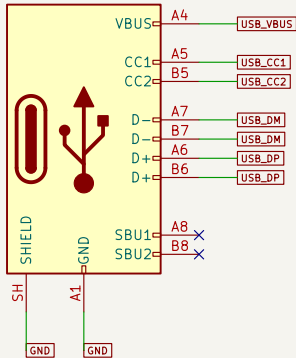
C69
100u / 10 V polymer, J3 bulk



C70
100n / 16 V X7R, SS soft-start



J4
USB_C_PROGRAM_POWER



R4
5.1k / 1%



R5
5.1k / 1%



U25 TPS2121 PRIORITY POWER MUX (TI SLVSEA3F). IN1 = 5V_IN is the priority source, IN2 = USB_VBUS the fallback; OUT = V5_SYS. Reverse-current blocking is always on for BOTH channels (Sec 9.3.6, IRCB 0.2/1/2 A, tRCB 10 us, VRCB 0/25/50 mV), so neither input can be back-fed from V5_SYS or from the other input. RON 56 mOhm typ / 100 mOhm max over temperature (Sec 7.5, VINx >= 5 V). PR1 (Sec 10.2.4.1 Eq.5, VPR1 = VIN1 x Rb/(Rt+Rb) compared with VREF): R42 32.4k / R43 10k -> IN1 released when 5V_IN falls to 1.04 V / 0.23585 = 4.410 V typ and re-selected at 1.06 V / 0.23585 = 4.494 V typ (4.20–4.62 V / 4.28–4.66 V over the VREF spec). The divider is centred on the RISING edge, not the falling one: the binding corner is a 5V_IN at its 4.75 V minimum meeting a VREF at its 1.10 V maximum, and 4.664 V worst-case rising keeps IN1 selected everywhere inside the declared 4.75–5.5 V window. OV1 (Sec 9.3.5 / 10.2.4.2 Eq.6, same divider form): R44 45.3k / R45 10k -> IN1 rejected at 1.06 V / 0.180832 = 5.862 V typ (5.585–6.083 V over the VREF spec, so it can never trip inside the declared 4.75–5.5 V input window). OV2 is tied to GND: the USB channel has no overvoltage cut-off ("Connect to GND if not required", Sec 6). CP2 is tied to GND, which selects the internal-VREF priority scheme of Table 9–3 – that is the mode PR1–vs–VREF describes, and it costs the TPS2121 fast-switchover path: the applicable spec is tSW = 100 us typ (Sec 7.5), not the 5 us tFSW that needs CP2 >= VREF. ILM (Sec 9.3.2 Eq.2, ILM = 65.2 / RILM'0.861 with RILM in kOhm, valid 18–100 kOhm): R46 80.6k -> 1.49 A typ; the datasheet characterises RILM = 80k as 1.0 / 1.5 / 2.0 A min/typ/max. Fast-trip OCP is 2.4 x ILM (Sec 9.3.3). SS (Sec 9.3.1 / Table 9–1): C70 100 nF -> 780 V/s at 5 V, so the ~240 uF on V5_SYS draws ~190 mA of inrush and OUT ramps in ~6.4 ms. ST (Sec 6, Sec 10.2.3) is an open-drain status output; HIGH when IN1 (or neither input) drives OUT, LOW when IN2 does, R47 10k to V3V3_SYS is inside the RST = 6–20 kOhm recommended operating range of Sec 7.3; PWR_SRC_ST lands on RP2354B GPIO39 (pin 48). Sec 11 gives no numeric minimum COUT, only "increase the capacitance on OUT to avoid output voltage drop"; C19 220 uF + C8/C9 2 x 10 uF on POWER 2 already exceed the 100–200 uF the datasheet design examples use, so no extra output capacitor is added here.

J3 INPUT NETWORK: C16 100 nF + C67/C68 2 x 4.7 uF X7R + C69 100 uF polymer, and D23 SMAJ5.0A (DO–214AC/SMA) clamping 5V_IN to GND. D23 is a UNI-directional TVS: its KiCad symbol inherits the generic A1/A2 pin names from the SM6T family, but the graphic and the DO–214AC band both put the CATHODE on pin 1, so pin 1 goes to 5V_IN and pin 2 (anode) to GND. 5.0 V standoff / 6.4 V VBR min sits above the 5.5 V input maximum and below the TPS2121 5.585 V worst-case OV1 trip, so the clamp and the mux do not fight. J3 is XT30PW–M: pad 1 is the "–" terminal and pad 2 the "+" terminal in the KiCad footprint – see the comment in tools/build_power.py.

USB_VBUS feeds U25 IN2 directly; no inrush/current limiting on this sheet.
Data ESD protection is on POWER 2. No servo rail connection.

C7
1u / 10 V



Sheet: /POWER 1 / 5 V in, USB, priority power mux/
File: power_sources.kicad_sch

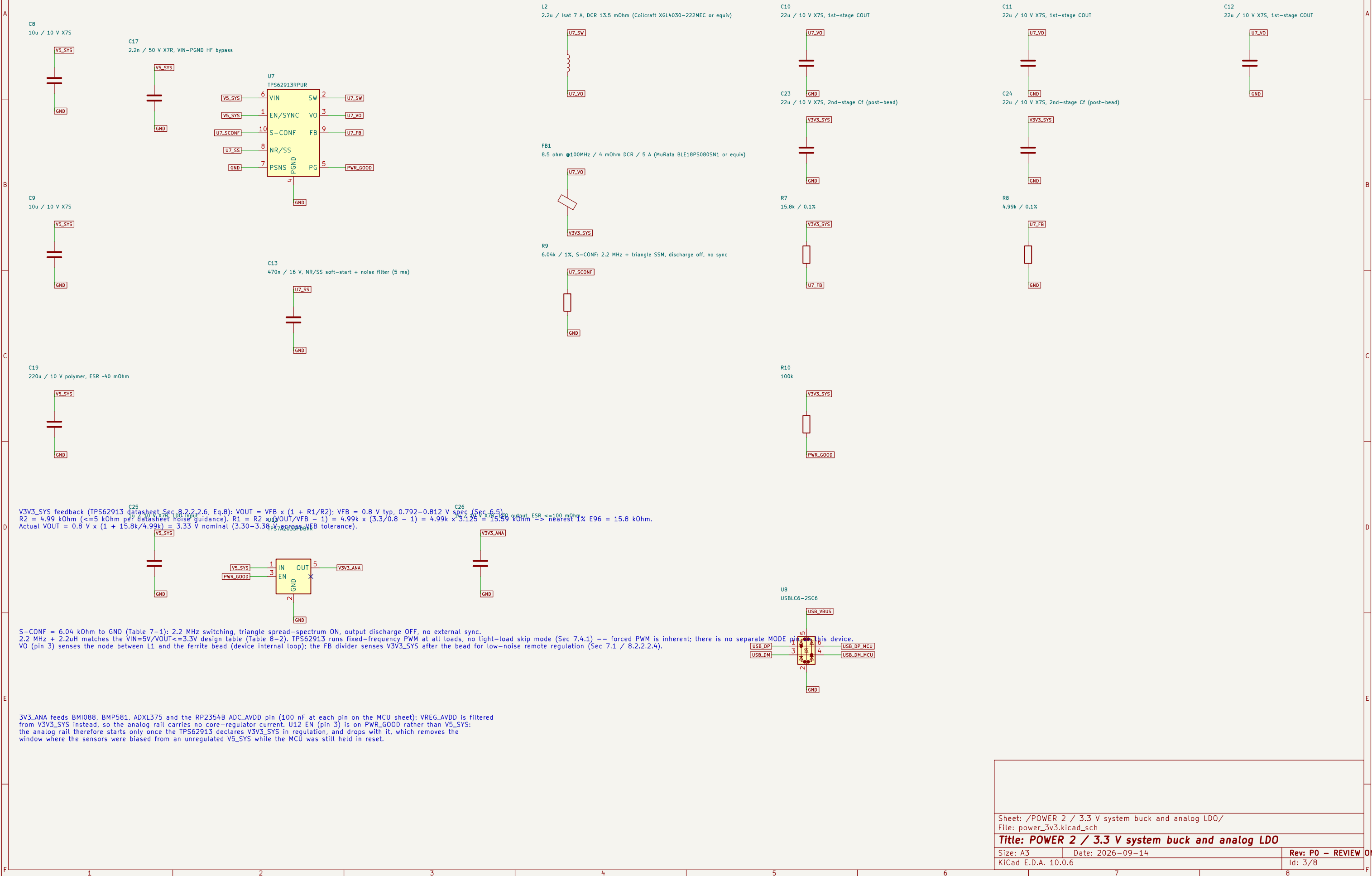
Title: POWER 1 / 5 V in, USB, priority power mux

Size: A3
KiCad E.D.A. 10.0.6

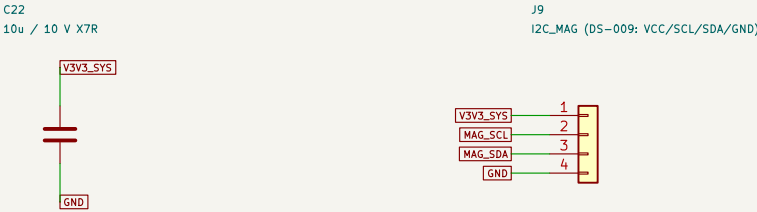
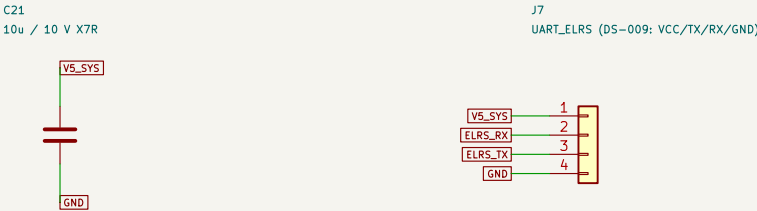
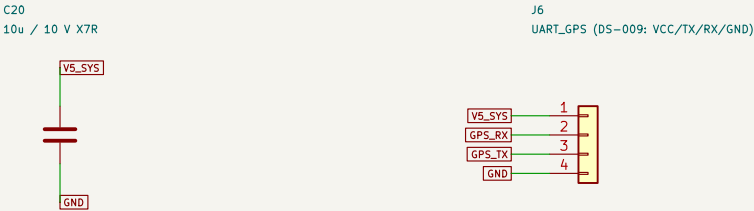
Date: 2026–09–14

Rev: P0 – REVIEW ONLY
Id: 2/8

AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.

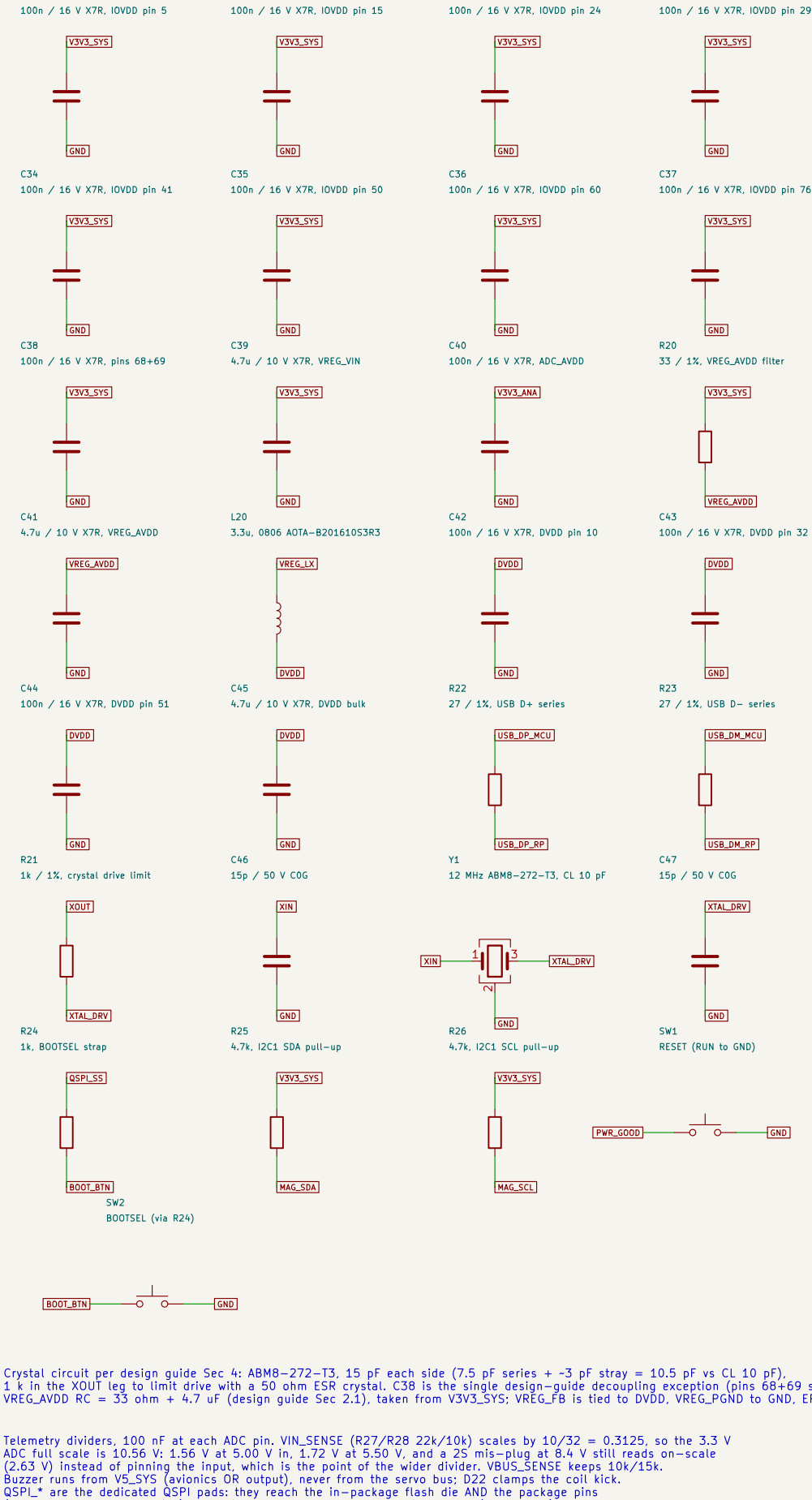


PIXHAWK DS-009 PIN ORDER. UART ports (J6, J7): 1 VCC, 2 TX, 3 RX, 4 GND. I2C port (J9): 1 VCC, 2 SCL, 3 SDA, 4 GND. Pin 2 of a UART port is the BOARD TX, i.e. the RP2354B UART TX output, which lands on the module RX input – that is the GPS_RX / ELRS_RX net, because a port label is named for the peripheral pin it belongs to (see the MCU sheet). Pin 3 is the board RX, fed by the module TX: GPS_TX / ELRS_TX.



RAILS: J6 and J7 VCC are on V5_SYS – GNSS and ELRS modules are 5 V–powered with their own on–module regulators, and their C20 / C21 10 uF local bulk moved to V5_SYS with them. Their SIGNAL pins stay 3.3 V CMOS straight from the RP2354B, which is what both module families expect. J9 VCC stays on V3V3_SYS: the magnetometer bus is 3.3 V and its pull–ups (R25/R26 on the MCU sheet) bias to V3V3_SYS, so a 5 V port pin there would fight them. J9 pin 2 = MAG_SCL, pin 3 = MAG_SDA.

RP2354B, QFN-80, 2 MB in-package QSPI flash. Digital rails on V3V3_SYS, core DVDD from the on-chip buck.
RAIL SPLIT: V3V3_ANA (TPS7A20 LDO) now feeds exactly one RP2354B pin – ADC_AVDD (pin 59, with C40). The core regulator input filter R20/C41 that produces VREG_AVDD is fed from V3V3_SYS instead, so the buck-converter reference current no longer flows out of the low-noise analog rail and the sensors do not share a rail with it; V3V3_ANA is left carrying only ADC_AVDD and the MEMS. Decoupling, core regulator, crystal and BOOTSEL follow the RP2350 hardware design guide; every pin net is a global label.



Crystal circuit per design guide Sec 4: ABM8-272-T3, 15 pF each side (7.5 pF series + ~3 pF stray = 10.5 pF vs CL 10 pF), 1 k in the XOUT leg to limit drive with a 50 ohm ESR crystal. C38 is the single design-guide decoupling exception (pins 68+69 share it). VREG_AVDD RC = 33 ohm + 4.7 uF (design guide Sec 2.1), taken from V3V3_SYS; VREG_FB is tied to DVDD, VREG_PGND to GND, EP to GND.

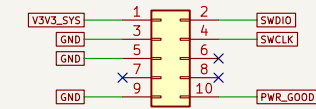
Telemetry dividers, 100 nF at each ADC pin. VIN_SENSE (R27/R28 22k/10k) scales by 10/32 = 0.3125, so the 3.3 V ADC full scale is 10.56 V: 1.56 V at 5.00 V in, 1.72 V at 5.50 V, and a 2S mis-plug at 8.4 V still reads on-scale (2.63 V) instead of pinning the input, which is the point of the wider divider. VBUS_SENSE keeps 10k/15k. Buzzer runs from V5_SYS (avionics OR output), never from the servo bus; D22 clamps the coil kick. QSP*_ are the dedicated QSPI pads: they reach the in-package flash die AND the package pins (RP2350 datasheet Sec 14.3), so U24 shares the bus with GPIO0/QMI CS1n (FLASH_CS1) as its select. RP2354 requires QSPI_IOVDD = 3.3 V, and IOVDD = 3.3 V to run a second QSPI device (Sec 14.9).

PORT NAMING: a port label is named for the peripheral pin it comes from, so GPS_TX / ELRS_RX are module outputs and land on MCU UART RX pins, while GPS_RX / ELRS_RX are module inputs fed by the MCU UART TX pins. On J6/J7 POWER 3, Pixhawk DS-009 order 1 VCC / 2 TX / 3 RX / 4 GND) pin 2 is therefore GPS_RX / ELRS_RX. J6 and J7 VCC are on V5_SYS; J9 VCC stays on V3V3_SYS. All port SIGNAL pins remain 3.3 V CMOS. GPIO39 (pin 48) is no longer spare: it reads PWR_SRC_ST, the open-drain ST output of U25 (TPS2121), which is high while 5V_IN (IN1) powers V5_SYS and low while USB VBUS (IN2) does.

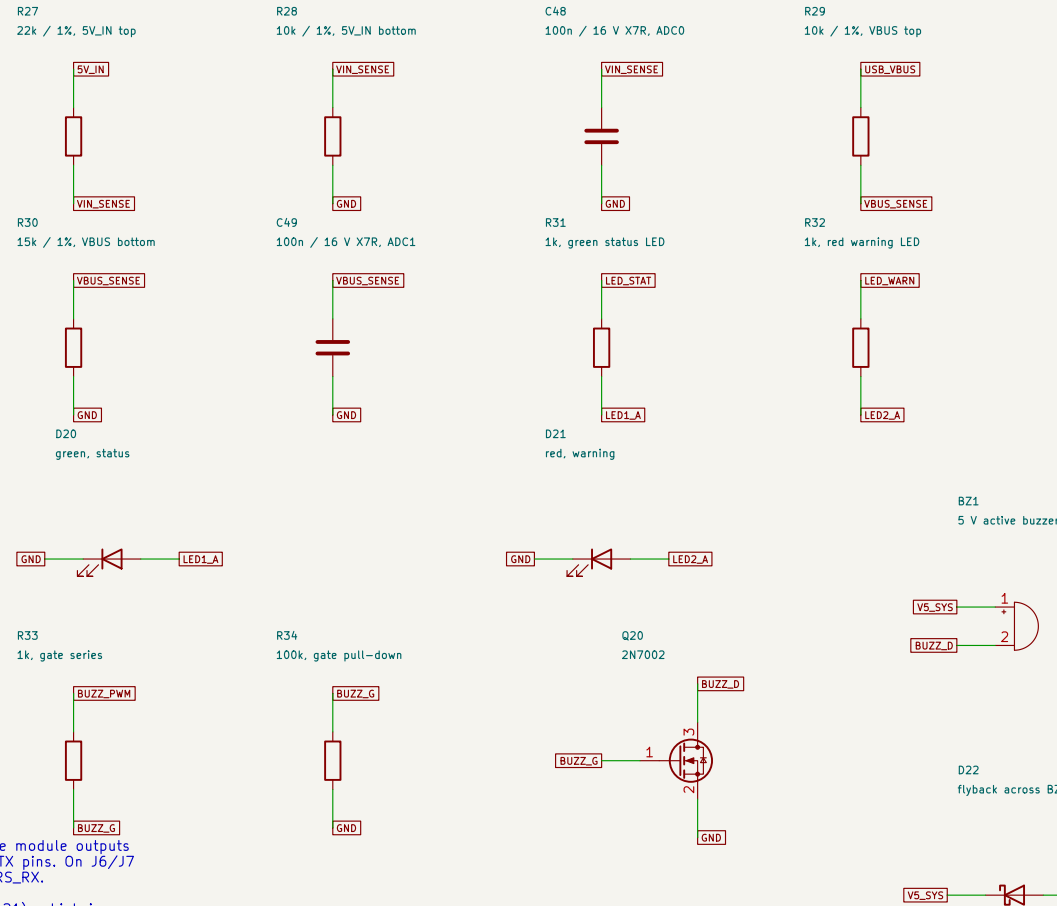
#FLG5 PWR_FLAG #FLG6 PWR_FLAG

GPIO MAP (pin = QFN-80 pin)			
GPIO0	77	FLASH_CS1	QMI CS1n, log flash select
GPIO1	78	(no connect)	spare
GPIO2	79	PWM1	PWM1 A
GPIO3	80	PWM2	PWM1 B
GPIO4	1	PWM3	PWM2 A
GPIO5	2	PWM4	PWM2 B
GPIO6	3	MAG_SDA	I2C1 SDA
GPIO7	4	MAG_SCL	I2C1 SCL
GPIO8	6	ELRS_RX	UART1 TX -> radio RX
GPIO9	7	ELRS_TX	UART1 RX <- radio TX
GPIO10	8	PWM5	PWM5 A
GPIO11	9	PWM6	PWM5 B
GPIO12	11	GPS_RX	UART0 TX -> GPS RX
GPIO13	12	GPS_TX	UART0 RX <- GPS TX
GPIO14	13	PWM7	PWM7 A
GPIO15	14	PWM8	PWM7 B
GPIO16	16	SENS_MISO	SPI0 RX
GPIO17	17	HG_ACC_CS	ADXL375 CS
GPIO18	18	SENS_SCK	SPI0 SCK
GPIO19	19	SENS_MOSI	SPI0 TX
GPIO20	20	IMU_ACC_CS	BMI088 CSB1
GPIO21	21	IMU_GYR_CS	BMI088 CSB2
GPIO22	22	BARO_CS	BMP581 CSB
GPIO23	23	IMU_ACC_INT	BMI088 INT1

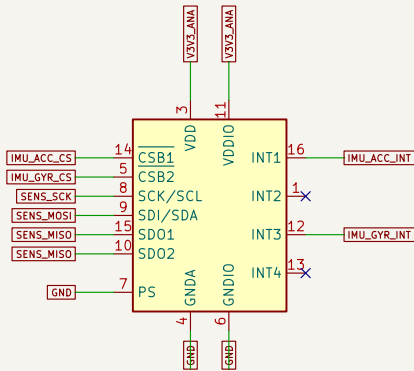
J10 SWD, Cortex-M 10-pin 1.27 mm



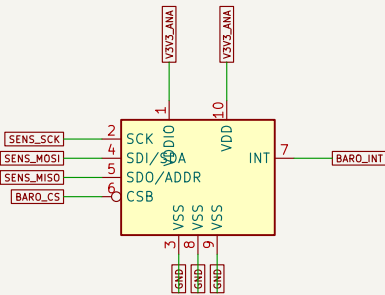
J10 pin 1 is 3V3 OUT for debugger reference only – never power the board from it. Pin 10 is the debugger reset line on PWR_GOOD/RUN: pins 6-8 (SWD, KEY, NC) unused. RUN is held by the TPS62913 power-good pull-up on POWER 2, so the MCU cannot run before V3V3_SYS is in regulation.



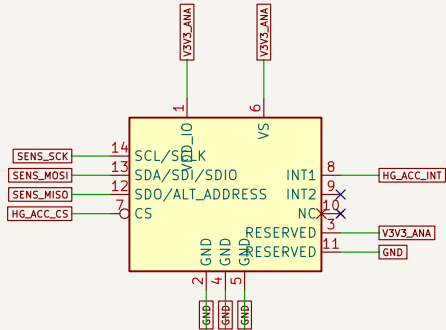
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TPS7A20 LDO) with 100 nF + 1 uF at every supply pin per datasheet.



U21
BMI088 6-axis IMU



U22
BMP581 barometer



U23
ADXL375 high-g accelerometer

C50
100n / 16 V X7R, U21 VDD



C52
100n / 16 V X7R, U21 VDDIO



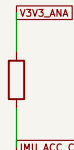
C51
1u / 10 V X7R, U21 VDD



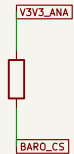
C53
1u / 10 V X7R, U21 VDDIO



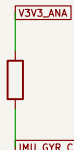
R48
10k, IMU_ACC_CS pull-up



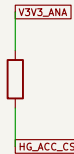
R50
10k, BARO_CS pull-up



R49
10k, IMU_GYR_CS pull-up



R51
10k, HG_ACC_CS pull-up



C54
100n / 16 V X7R, U22 VDD



C56
100n / 16 V X7R, U22 VDDIO



C55
1u / 10 V X7R, U22 VDD



C57
1u / 10 V X7R, U22 VDDIO



C58
100n / 16 V X7R, U23 VS



C60
100n / 16 V X7R, U23 VDDIO



C59
1u / 10 V X7R, U23 VS



C61
1u / 10 V X7R, U23 VDDIO



U21 BMI088 (datasheet Sec 6.2): PS tied to GND selects SPI, CSB1 = accelerometer (IMU_ACC_CS), CSB2 = gyroscope (IMU_GYR_CS); SD01 is the accelerometer data output and SD02 the gyroscope one, both high-Z while deselected, so they share SENS_MISO (the symbol is Sensor_Motion:BMI088 with SD02 retyped passive so ERC accepts that). INT1 = accelerometer interrupt, INT3 = gyroscope interrupt; INT2 and INT4 are unused and left open. VDD and VDDIO are both on V3V3_ANA; GND_A and GNDIO are the same board ground.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT, VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT.
Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48–R51: 10k pull-ups to V3V3_ANA on the four SPI chip selects (IMU_ACC_CS, IMU_GYR_CS, BARO_CS, HG_ACC_CS). Same rationale as R35 on FLASH_CS1 (BOARD 3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus or leave two SDO drivers on SENS_MISO. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

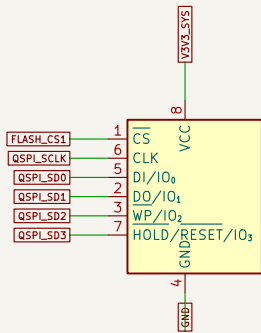
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-14

Rev: P0 – REVIEW ONLY

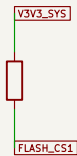
Id: 6/8

Log flash on the shared QSPI bus with GPIO0/QMI CS1n as its chip select, and a latched microSD wired for 4-bit PIO (also SPI-compatible). Both on V3V3_SYS with local bulk, per the rail split in DESIGN_SPEC.



U24
W25Q128JVSIQ, 128 Mbit QSPI NOR

R35
10k, FLASH_CS1 pull-up



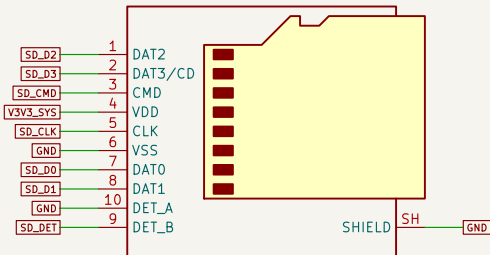
C62
100n / 16 V X7R, U24 VCC



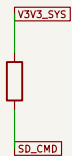
C63
1u / 10 V X7R, U24 VCC



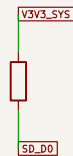
J11
microSD, Molex 104031-0811 push-push



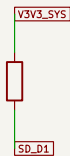
R36
10k, SD_CMD pull-up



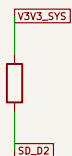
R37
10k, SD_D0 pull-up



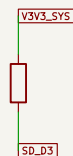
R38
10k, SD_D1 pull-up



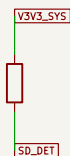
R39
10k, SD_D2 pull-up



R40
10k, SD_D3 pull-up



R41
10k, SD_DET pull-up



C64
100n / 16 V X7R, at the socket



C65
10u / 10 V X7R, at the socket



C66
47u / 6.3 V X5R, at the socket



U24 shares the dedicated QSPI pads with the RP2354B in-package flash die (datasheet Sec 14.3): CLK = QSPLSCLK, D1/IO0 = QSPLSD0, D0/IO1 = QSPLSD1, WP/IO2 = QSPLSD2, HOLD/IO3 = QSPLSD3. Only the chip select differs – FLASH_CS1 comes from GPIO0 (QMI CS1n), pulled up to V3V3_SYS by R35 so the part is deselected before the MCU drives it. QSPI_SS (the internal die) keeps its own strap on the MCU sheet.

J11: DET_A to GND and DET_B to SD_DET with a 10k pull-up, so the input reads low with a card inserted. The shield goes to GND. 100 nF + 10 uF + 47 uF sit at the socket (DESIGN_SPEC "Rails": SD transients stay on the buck rail), 10k pull-ups on CMD and DAT0–DAT3 are the SD-standard idle bias and also keep DAT3/CD high for 4-bit mode. VDD = V3V3_SYS.

Sheet: /BOARD 3 / log flash and microSD/
File: storage.kicad_sch

Title: BOARD 3 / log flash and microSD

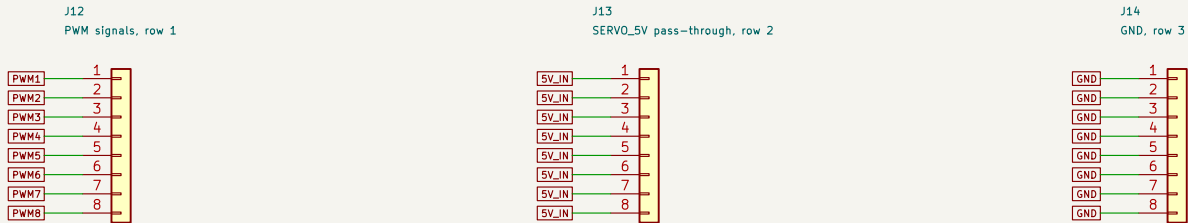
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-14

Rev: P0 – REVIEW ONLY

Id: 7/8

Eight PWM outputs on a 3-row 2.54 mm block: signal /2SERVO_5V / GND. No series resistors –3.3 V logic straight from the RP2354B, which is what servos and ESCs expect.



3.3 V logic; servo/ESC power is the external BEC bus, 3 A limit is the BEC’s.

Row 2 is SERVO_5V = the 5V_IN net, i.e. the J3 input BEFORE U25 (the TPS2121 priority power mux). Servo and ESC current therefore never crosses the board rails, V5_SYS or the mux; in particular it is not counted against the mux ILM current limit (1.49 A, set by R46). It returns through the row 3 ground pins.
Nothing on this sheet loads V3V3_SYS or USB_VBUS.

No 3-row generic connector symbol or 2.54 mm 3x08 footprint ships with this KiCad install, so the header is drawn as three stacked Conn_01x08 symbols (J12 signal, J13 SERVO_5V, J14 GND) with three 1x08 footprints. Place them on 2.54 mm centres to form the usual 3x8 servo block; column n is J12 pin n / J13 pin n / J14 pin n.

PWM MAP
PWM1 = GPIO2 PWM5 = GPIO10
PWM2 = GPIO3 PWM6 = GPIO11
PWM3 = GPIO4 PWM7 = GPIO14
PWM4 = GPIO5 PWM8 = GPIO15